

V1 Business Architecture

Document Metadata

Field	Value
Document	V1 Business Architecture
Product	Yoga Center Attendance App
Version	V1
Status	Completed / Baseline
Purpose	Business architecture reference for V2 planning
Source	Existing V1 implementation
Baseline	Repository <code>main</code> at commit <code>75b1664</code> (1 Oct 2026), database migrations <code>0001 – 0024</code>
Audience	Product, design and development teams

How to read this document

This document records what V1 **actually does**. Where the written product documentation and the implementation disagree, the implemented behaviour is recorded and the disagreement is listed in Appendix A. Nothing here is a proposal.

Each rule carries an evidence marker:

Marker	Meaning
DB	Enforced by the database (constraint, trigger, row-level security, or database function). Holds no matter which screen is used.
App	Enforced by the application's server-side logic.
UI	Behaviour of the screens only (e.g. what options are offered).

Marker	Meaning
Test	Covered by the automated unit test suite (370 tests, all passing at baseline).
Doc	Stated in the V1 product/UX documentation (docs/01-product.md , docs/02-ux.md).
Not confirmed	No evidence of the behaviour, or evidence is ambiguous. Not assumed.

1. Executive Summary

V1 is a web application that runs **one yoga center**. Center staff use it to manage students, their memberships and batch enrollments, the center's recurring weekly class timetable, the dated class sessions that timetable produces, and Present/Absent attendance for each session. It provides attendance history, three attendance reports with CSV/Excel export, a role-specific dashboard, and center settings.

There are exactly **two roles**: **Admin** (full management) and **Instructor** (records and corrects attendance for their own sessions only). Students do not sign in.

The business architecture rests on five ideas, each enforced in the database rather than only in screens:

1. **Eligibility is computed, not chosen.** A student appears on a session's roster only if, *on that session's own date*, they are an active student, hold an active enrollment in the batch, are assigned to that specific recurring class, and hold a membership covering the date.
 2. **History is frozen at the moment it is recorded.** When attendance is first saved, the session's details and its eligible-student list are captured permanently. Later changes to students, enrollments, memberships, batches or schedules cannot rewrite it.
 3. **Nothing historical is deleted.** Students, batches, instructors, enrollments, memberships, sessions and attendance are deactivated, cancelled, closed or corrected — never removed. The only deletions allowed are for records that have never become part of history (an unused schedule; a schedule assignment that has not started).
 4. **Schedules are versioned, sessions are projected.** Editing the timetable closes the old version and starts a new one. Dated sessions are calculated from the timetable and only stored when something must attach to them.
 5. **One center, one clock.** Every "today", every session status and every membership status is evaluated in the center's configured time zone.
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2. Product Scope

2.1 Purpose

Make daily attendance fast while giving the center a reliable way to manage changing batches, schedules, enrollments and membership periods without damaging historical attendance. (*Doc*)

2.2 Product areas implemented in V1

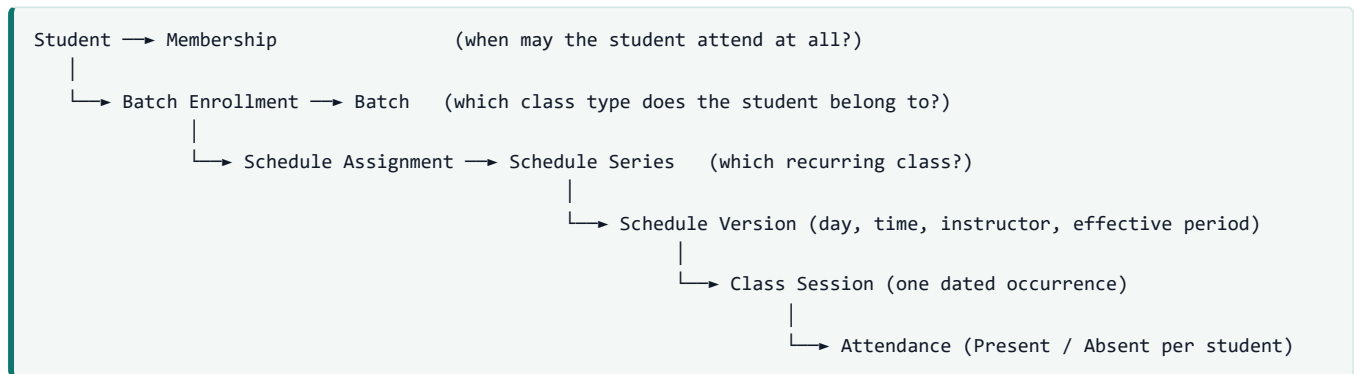
Area	Admin	Instructor	Evidence
Dashboard	Yes	Yes (own classes only)	App, DB
Students (incl. enrollments)	Yes	No	App, DB
Memberships (incl. receipt)	Yes	No	App, DB
Batches	Yes	No	App, DB
Schedule	Yes	No	App, DB
Attendance (sessions, Take Attendance)	Yes	Yes (own sessions only)	App, DB
Attendance History	Yes	Yes (own sessions only)	App, DB
Reports (+ CSV / Excel export)	Yes	No	App
Settings (Center Profile, Instructors, Roles & Permissions)	Yes	No	App, DB
Assigned Classes	—	Not built (listed in UX navigation; deferred)	App

2.3 Explicitly not in V1

See Section 24.

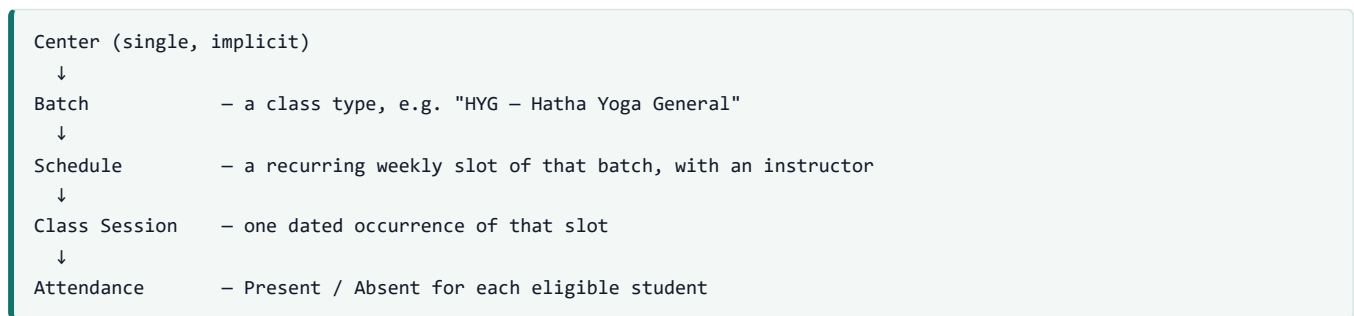
3. V1 Business Model

3.1 Core model



(DB, Doc)

3.2 Center-level operating model



The center is not a record that other records point to. It is a single settings record (name, logo, contact details, time zone, currency). Every other record implicitly belongs to the one center. (DB)

3.3 Distinct concepts

V1 keeps these concepts separate on purpose (Doc, DB):

Concept	What it answers
Student	Who is this person?
Membership	Over which dates is the student allowed to attend? Was it paid?
Batch	Which class type?
Batch Enrollment	Over which dates does the student belong to this batch?
Schedule Assignment	Which of the batch's recurring classes does the student attend, and since when?
Schedule (series + versions)	When does a recurring class run, who teaches it, from when to when?
Class Session	What happened (or will happen) on a specific date?
Attendance	Was this eligible student Present or Absent at that session?

Membership and enrollment are **independent**. One membership covers every batch the student is enrolled in during its dates; there is no stored link between a membership and an enrollment. (DB, Doc)

4. User Roles

Role	Description	How the account is created	Evidence
Admin	Center owner / manager. Full management access.	Out-of-band only. Every new account is created as Instructor; promotion to Admin is done directly in the database. No screen creates or promotes an admin.	DB
Instructor	Teaches classes; records and corrects attendance for their own sessions.	Admin creates an Instructor record, then uses Provide Login Access , which sends an email invitation.	App, DB
Student	Managed record only. No login.	Created by Admin.	Doc, DB

There are no custom roles, no per-role configuration and no third staff role. The Settings → Roles & Permissions screen is **read-only**: it displays the fixed access model, derived from the same source that controls navigation. (*App*)

5. Role Responsibilities and Permissions

5.1 Permission matrix

Capability	Admin	Instructor	Enforced by
Sign in, reset own password via email	Yes	Yes	App
Change own password inside the app (Settings)	Yes	No (Settings is admin-only; instructors use Forgot Password)	App
View dashboard	All classes + center KPIs + recent activity	Own classes only	App, DB
Students: view, add, edit, activate / deactivate	Yes	No	App, DB
Enrollments & schedule assignments	Yes	No	App, DB
Memberships: add, edit, renew, cancel, receipt	Yes	No	App, DB
Batches: add, edit, activate / deactivate	Yes	No	App, DB
Schedules: add, edit (version / direct), deactivate, delete unused	Yes	No	App, DB
View session lists and session details	All sessions	Own sessions only	App, DB
Edit a single session (instructor / time)	Yes	No	App
Mark a session Cancelled / Holiday	Yes	No	App
Take attendance	Any session	Own sessions only	App, DB
Edit (correct) saved attendance	Any session	Own sessions only	App, DB
View attendance history	All	Own sessions only	App, DB
Reports & export	Yes	No	App
Center profile, regional settings	Yes	No	App, DB
Manage instructors, provide login access	Yes	No	App, DB

5.2 What "own session" means for an instructor

- For a session that has been stored (materialized), the instructor named **on the session itself** owns it. If an Admin reassigns one session to a different instructor, ownership moves with it — the new instructor gains it, the original instructor loses it, even though the recurring schedule still names the original. *(DB)*
- For a session not yet stored, ownership comes from the schedule's current instructor. *(DB)*
- An instructor sees student names and phone numbers **only** as the eligible roster of a session they own. They cannot browse the student list, memberships or enrollments. *(DB)*

5.3 Deactivated or unlinked instructors

- A deactivated instructor, or a signed-in user not linked to any instructor record, receives **no data**: every instructor data permission resolves to nothing at the database. *(DB)*
- Such a user can still sign in and reach the instructor screens, which appear empty. A dedicated "your access is disabled" screen is **not built**. *(App)*

6. Student Management

Purpose. Maintain the center's students and give the Admin a single place to see each student's enrollments, membership and recent attendance.

Who. Admin only. (*App, DB*)

6.1 Student information

Field	Rule	Evidence
Student ID	System-generated, sequential YC-000001, unique, immutable . Any value supplied by a caller is overwritten; any attempt to change it is rejected.	DB
Full name	Required, ≤ 100 characters.	App
Phone	Required, digits only, ≤ 30. Stored with a separate country calling code (e.g. +91). Not unique .	App, DB
Email	Optional, valid format, ≤ 255, stored lower-case. Not unique .	App
Date of birth	Optional; cannot be in the future.	App
Gender	Optional: Male / Female / Other.	App, DB
Join date	Required.	App, DB
Profile photo	Optional; JPEG / PNG / WebP, ≤ 2 MB.	App, DB
Notes	Optional, ≤ 1000 characters.	App
Status	Active / Inactive. New students are Active.	DB

6.2 Actions

Action	Behaviour	Evidence
Add Student	Guided three-step flow: Student → Add Membership (skippable) → Add Batch Enrollment.	App, Doc
Edit Student	Edits details only; status is never changed by Edit.	App
Deactivate / Activate	Separate action with a Review → Confirm step. Retains every historical record. Reversible.	App, Doc
Delete	Not possible . No delete permission exists.	DB
Search / filter	Search by name, phone, or name/code of a batch the student is actively enrolled in. Filters: status, batch, membership (Active / Expired / None).	App

6.3 Business rules

- Deactivation does **not** cascade: it does not end enrollments, assignments or memberships. (*App*)
- An inactive student is excluded from attendance eligibility. Student status has no history — eligibility reads the student's **current** status (see Section 23.3 for the consequence). (*DB*)
- The Student Details page shows enrollments (with assigned schedules), the current membership, recent attendance, profile and notes. An enrollment with no schedule assignment is visibly flagged. (*App, Test*)

7. Instructor Management

Purpose. Maintain the people who teach. An instructor is an independent record, assigned to classes at the **schedule** level. (*Doc, DB*)

Who. Admin only (Settings → Instructors). (*App, DB*)

Field / action	Rule	Evidence
Full name	Required, ≤ 100.	App
Email	Optional; unique across instructors (case-insensitive). Required before login access can be granted.	DB, App
Phone	Optional, digits only, with country code.	App, DB
Photo	Optional; JPEG / PNG / WebP, ≤ 2 MB.	App, DB
Status	Active / Inactive; quick toggle from the list or on Edit. Reversible.	App
Delete	Not possible.	DB
Provide Login Access	Separate, explicit action. Preconditions: instructor is Active, has an email, and has no login yet. Sends an email invitation; the invited account is linked to this instructor record.	App
Revoke login access	Not built. Deactivating the instructor removes their data access instead.	App, DB
Assign to classes	Done by choosing the instructor on a schedule, or on a single session. Only Active instructors are offered.	UI

Rules.

- One sign-in account links to at most one instructor record, and vice versa. (*DB*)
 - The link is made from the account created by the invitation — never by matching email — so later email edits do not change who the instructor is. (*App*)
 - Removing a sign-in account never removes the instructor record. (*DB*)
 - Deactivating an instructor does **not** reassign or close their schedules; those classes continue to appear (see Section 23.4). (*App*)
-

8. Batch Management

Purpose. Represent the class types the center offers. A batch has no fixed time; its times come from its schedules. *(Doc)*

Who. Admin only. *(App, DB)*

Field	Rule	Evidence
Name	Required, ≤ 100.	App
Short code	Required, ≤ 20, stored upper-case, unique case-insensitively (the weekly timetable identifies a batch by code alone).	DB, App
Category	Optional, ≤ 50.	App
Description	Optional, ≤ 500.	App
Colour	Required, one of 10 palette colours (default teal). Visual identity only.	DB
Image	Optional; JPEG / PNG / WebP, ≤ 2 MB.	App, DB
Status	Active / Inactive, changed through Edit Batch.	App
Capacity	Not supported in V1.	Doc

Displayed batch status (Batches list) is derived, not stored *(App, Test)*:

Displayed	Condition
Inactive	Stored status is Inactive.
Active	Stored Active, and either no schedules at all or at least one schedule currently in effect.
Upcoming	Stored Active, no current schedule, at least one active schedule starting in the future.
Completed	Stored Active, and all of its schedules have ended or been deactivated.

Rules.

- A batch can have many schedules, including several on the same weekday. *(Doc, DB)*
 - Batches are never deleted; history remains available after deactivation. *(DB)*
 - Deactivating a batch does **not** deactivate its schedules or end its enrollments. Inactive batches are no longer offered when creating a new enrollment. *(App, UI)*
 - Batch Details has Overview, Students (every enrollment, active and historical, with assigned schedules), Schedules, and Attendance (history scoped to the batch). *(App)*
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9. Enrollment

V1 has two levels of enrollment: the **batch enrollment** and the **schedule assignment** that sits inside it.

```
Student
↓
Batch Enrollment (student ↔ batch, with effective start / optional end, Active / Inactive)
↓
Schedule Assignment(s) (which recurring class(es) of the batch, with effective start / optional end)
↓
Batch / Schedule Series
```

Who. Admin only, from Student Details. (*App, DB*)

9.1 Batch enrollment rules

Rule	Evidence
A student may be enrolled in several batches at the same time (e.g. combo offers).	Doc, DB
At most one Active enrollment per student per batch . Inactive (historical) enrollments in the same batch are kept, so a student can re-enroll later.	DB
Effective start date required; end date optional and not before start.	DB, App
New enrollments are created Active. Status can be changed on Edit.	App
Only Active batches are offered when choosing a batch.	UI
Enrollments are never deleted.	DB
Enrollment start / end dates are freely editable, including to past dates.	App

9.2 Schedule assignment rules

Rule	Evidence
Being enrolled in a batch does not by itself make a student eligible for the batch's classes; they must be assigned to the specific recurring class.	DB, Doc
At least one schedule must be selected when adding or editing an enrollment.	App
Every selected schedule must belong to the enrolled batch (checked on the server).	App
An enrollment may hold several assignments at once (e.g. both the 6 AM and the 7 PM class).	DB
Assignments point to the schedule series (the stable identity of a recurring class), so editing a schedule never detaches its students.	DB
No status field: an assignment applies exactly to the dates in its period.	DB
The same schedule cannot be assigned twice to one enrollment over overlapping dates.	DB
Overlapping class times across a student's own assignments are not blocked.	Doc, DB
New enrollment: every selected schedule starts on the enrollment's start date.	App
Changing the selection on Edit requires a separate schedule change date , which cannot be before today. Removed schedules are closed the day before that date; added schedules start on it; unchanged ones are left alone.	App
A removed assignment that has not started yet (starts on or after the change date) is withdrawn (deleted) rather than closed. The database permits deleting only assignments that start today or later.	DB, App

9.3 Relationships

- Enrollment drives eligibility (Section 14.2) and the "covered batch enrollments" shown on a membership. *(App)*
 - Enrollment and its assignments are saved in separate steps, not one transaction. If assignment saving fails, the enrollment exists without a schedule and is visibly flagged on Student Details and Batch Details. *(App)*
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10. Membership Management

Purpose. Record the period during which a student may attend, the price and whether it has been paid. Membership is one of the four eligibility conditions. (*Doc, DB*)

Who. Admin only. (*App, DB*)

10.1 Membership information

Field	Rule	Evidence
Membership ID	System-generated, sequential MEM-000001, unique, immutable.	DB
Student	Required. The standalone Add Membership flow offers active students only; whether a membership can be added from an inactive student's own page is Not confirmed as intended (no status check exists).	DB, UI
Plan	Monthly / Quarterly / Custom.	DB, App
Start date	Required.	DB
End date	Required; not before start; same-day allowed. Auto-calculated (Monthly = start + 1 month – 1 day; Quarterly = start + 3 months – 1 day; Custom = manual) and always editable.	DB, App
Amount	Required, greater than 0.	DB, App
Currency	Stamped from the center currency when the membership is created; never converted afterwards.	DB, App
Payment status	Paid / Pending; defaults to Pending.	DB, App
Notes	Optional, ≤ 1000.	App
Cancelled at	Set once by Cancel; never cleared.	DB

10.2 Membership status (derived, never stored)

Status	Condition (evaluated on the center's current date)
Cancelled	Has been cancelled — overrides everything else.
Upcoming	Today is before the start date.
Active	Today is between start and end, both inclusive.
Expired	Today is after the end date.

A further list-level indicator, **Expiring Soon**, marks an Active membership with 7 or fewer days left. (*App, Test*)

10.3 Actions

Action	Behaviour	Evidence
Add	From the Memberships list (select student), from Student Details, or within Add Student.	App
Edit	Plain save of all editable fields. Available for every membership, including cancelled ones.	App
Renew	Always creates a new membership; the previous one is untouched. Defaults: start = previous end + 1 day, same plan; amount carried over only if the previous one was priced in the current center currency, otherwise left blank.	App, Doc

Action	Behaviour	Evidence
Cancel	Review → Confirm. Record retained. Cannot be cancelled twice. Cannot be undone. Does not affect enrollments.	App, DB
Delete	Not possible.	DB
Receipt	Printable receipt generated on demand from the membership, the student and the center profile (nothing stored). Includes a click-to-chat WhatsApp link pre-filled with the receipt details — a plain link, not an integration.	App, Doc

10.4 Business rules

- **No overlapping memberships.** A student cannot hold two non-cancelled memberships whose dates overlap (inclusive). Cancelled memberships are ignored by this rule. Enforced in the database, so it holds under concurrent use. *(DB, App)*
 - **Payment status does not affect membership status or eligibility.** A Pending membership is fully valid for attendance. *(Doc, App)*
 - **Cancellation only affects the future.** For eligibility, a membership counts on session dates strictly **before** the center-calendar day on which it was cancelled. Cancelling today never removes eligibility from a session that already happened. *(DB, Test)*
 - Expired and cancelled memberships remain in history. *(DB)*
 - Covered batch enrollments are derived from date overlap; nothing is stored. *(App)*
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11. Payment Management

V1 has **no payment entity**. Payment is recorded only as an attribute of the membership.

Student
↓
Membership — Amount, Currency, Payment Status (Paid / Pending)

Aspect	V1 behaviour	Evidence
Payment record / transaction	Not supported. No separate payment table, no payment date, no method, no partial payments.	DB
Payment status	Paid / Pending on the membership; editable via Edit Membership.	DB, App
Effect on eligibility	None.	Doc, App
Online payment gateway	Not supported.	Doc
Automatic billing / reminders	Not supported.	Doc
Tax, discount, invoice numbering	Not supported. The membership ID acts as the receipt number.	App
Revenue reporting	Not supported.	Doc
Filtering	Memberships list can be filtered by payment status.	App

The relationship "Student → Membership → Payment" therefore exists in V1 only as **Student → Membership (with payment status)**.

12. Schedule Management

Purpose. Define when each batch meets, who teaches, and from when to when, so that class sessions can be identified without manual entry. (*Doc*)

Who. Admin only. (*App, DB*)

12.1 Schedule information

Field	Rule	Evidence
Batch	Required; cannot be changed after creation.	DB, App
Instructor	Required; only Active instructors offered.	DB, UI
Day of week	One weekday per schedule.	DB
Start / end time	End must be after start. End defaults to start + 60 minutes and is editable.	DB, App
Effective from	Required.	DB
Effective until	Optional (open-ended); not before effective from.	DB
Status	Active / Inactive.	DB

12.2 Schedule series and versions

```
Schedule Series ("the Monday 6:00 AM Hatha Yoga General class")
├─ Version 1 effective 01 Jan - 31 Mar (instructor A, 6:00-7:00)
└─ Version 2 effective 01 Apr - open (instructor B, 6:00-7:00)
```

- Every schedule belongs to a series. Editing never creates a new series. (*DB*)
- Students are assigned to the series; sessions remember the exact version they came from. (*DB, Doc*)

12.3 Creating

Add Schedule accepts several weekdays and several time slots at once. Every day × time-slot combination becomes its own independent schedule (and its own series). (*App*)

12.4 Editing — three paths

Situation	Behaviour	Evidence
Unused schedule (no stored session of any status, only one version, no assignment that has started)	Corrected in place : day, time, instructor and dates may all change; no effective date needed. Students with not-yet-started assignments move with it.	DB, App, Test
Used, and the new effective date is after the current version's start	Current version closed the day before; a new version starts on the chosen date.	App
Used, but the new effective date is on or before the current version's start (not started yet)	Updated in place.	App
New effective date before today	Rejected.	App

A used schedule never changes its weekday; ticking extra days on Edit creates additional new schedules. (*App*)

12.5 Deactivating and deleting

Action	Behaviour	Evidence
Deactivate	Sets an effective-until date (defaults to today, editable, must not be before effective from) and marks the schedule Inactive. Cannot be repeated. No reactivate — a new schedule is added instead.	App
Delete	Only an unused schedule. Before confirming, the Admin sees how many students hold unstarted assignments and how many past dates were never recorded and will disappear. Unstarted assignments are withdrawn with it. The check is repeated inside the delete itself, so a session recorded meanwhile blocks it.	DB, App, Test

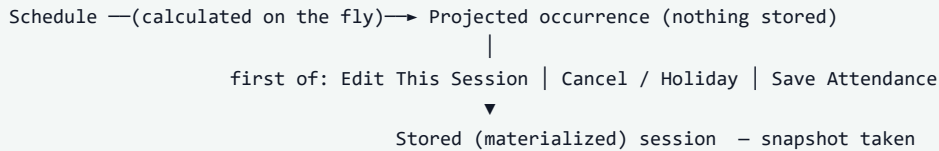
12.6 Rules

- Overlapping schedules (same batch or same instructor, same day and time) are **allowed** and shown side by side. *(Doc, DB)*
 - A schedule whose effective period has ended keeps its stored status; any "ended" indication is derived. *(Doc, App)*
 - Schedule changes never alter stored sessions (Section 13.4). *(DB)*
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13. Class Sessions

Purpose. A class session is one dated occurrence of a schedule — the record attendance attaches to. *(Doc)*

13.1 Projection and materialization



- Occurrences are **calculated** from schedules for display; nothing is stored just by looking. *(App, DB)*
- A session is **stored** only on its first session-specific edit, its first Cancel / Holiday, or its first attendance save. *(App, DB)*
- At most one stored session per schedule per date. *(DB)*
- There are **no ad-hoc sessions**: every session comes from a schedule. *(Doc, DB)*
- Sessions have **no product-facing ID**; they are addressed by schedule + date. *(Doc)*

13.2 Session status

Stored	Displayed
scheduled	Upcoming before start time; Ongoing from start through end time; Completed after end time (display only — nothing is written).
completed	Completed — only ever set by saving attendance.
cancelled	Cancelled
holiday	Holiday

All time comparisons use the center's time zone. *(DB, App, Test)*

13.3 Session actions

Action	Who	Allowed when	Changes	Evidence
Edit This Session	Admin	Displayed status is Upcoming	Instructor and/or start/end time of this one date. Date and batch cannot change. Writes nothing if nothing changed.	App, Test
Mark Cancelled / Holiday	Admin	Stored status is still scheduled (whatever it displays)	Status + optional note (≤ 500). Irreversible . Recurring schedule unaffected.	App
Take / save attendance	Admin; owning Instructor	Not cancelled/holiday; session date is today or earlier	See Section 14.	DB

Because saving attendance sets **completed**, a session with attendance can no longer be cancelled or marked holiday. *(App)*

13.4 Snapshot and historical integrity

When stored, a session captures its batch, instructor, date and times. From then on later schedule edits, versions or deactivation never change it; only Edit This Session can, and only while it is Upcoming. Stored sessions are never deleted. *(DB, Doc)*

13.5 Session lists

- **Today / All Sessions** combine stored and calculated occurrences without duplicates. All Sessions defaults to today through the next 90 days; filters: date range, batch, instructor, displayed status, search. (*App*)
 - Calculation is bounded by each schedule's effective dates, not its status. Deactivation works because it sets an end date. (*App*)
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14. Attendance Management

Purpose. Record, quickly, who was Present or Absent at each session. *(Doc)*

14.1 Statuses

Status	Stored?
Present	Yes
Absent	Yes
Unmarked	No — an eligible student with no record is unmarked.

There is no online/offline mode, late, excused or partial status. *(DB, Doc)*

14.2 Eligibility — who appears on the roster

A student is eligible for a session only if **all four** hold, evaluated on the **session's own date** (never "today") *(DB, Doc, Test)*:

#	Condition
1	The student is Active (current status).
2	The student has an Active enrollment in the session's batch whose dates cover the session date.
3	That enrollment has a schedule assignment to the session's schedule series covering the session date.
4	The student has a membership covering the session date that had not been cancelled before that date.

The batch used is the session's own (snapshot) batch. If a student has several qualifying memberships, the roster displays one (not cancelled preferred, then most recently started); this affects display only. *(DB)*

14.3 Saving attendance

```
Open session → roster of eligible students (all Unmarked initially)
  → mark individually / "Mark All Present" / change marks
  → Save
    | session stored if needed (snapshot of batch, instructor, times)
    | (first save only) eligible list frozen permanently
    | every submitted student re-checked as eligible – any failure rejects the whole save
    | Present / Absent upserted (one record per student per session)
    | session status → completed
```

All steps after the session is stored happen as **one atomic database operation**. *(DB)*

Rule	Evidence
Rejected for Cancelled or Holiday sessions.	DB
Rejected for a session dated after today (center time zone).	DB
No lower date limit: attendance can be taken for any past occurrence of a schedule.	DB, App
Saving with some or all students unmarked is allowed and still completes the session.	DB, Doc
A student may be marked only once per save; at most one record per student per session.	App, DB
Records are never deleted. A student left out of a later save keeps their existing mark.	DB
An instructor may save only for sessions whose stored instructor is them.	DB

14.4 Correcting attendance

- Admin can correct any completed session; an Instructor only their own. *(DB, App)*
- Corrections update the existing record in place; there is a Review Changes step before saving. *(App)*
- Corrections are validated against the **frozen** eligible list, so a student who has since left the batch can still be corrected. *(DB)*
- It is **not possible** to add a student who was not on the frozen list to a completed session (no mechanism exists; recorded as an undecided item). *(DB, Doc)*

14.5 Summary figures

Figure	Rule
Eligible	Size of the (frozen or live) eligible list.
Present / Absent	Count of records.
Unmarked	Eligible – all records, never below zero.
Attendance %	Present ÷ Eligible × 100 (not ÷ marked). Zero eligible shows 0% on session screens.

(App, DB)

15. Attendance History

Purpose. Review what already happened, read-only by default, with an explicit Edit for corrections. (*Doc*)

Aspect	V1 behaviour	Evidence
Who	Admin (all), Instructor (own sessions).	App, DB
What is listed	Only stored sessions with status completed . Cancelled, holiday and never-recorded occurrences do not appear.	App
Default range	No default limit (whole history).	App
Order	Newest first.	App
Filters	Date range, batch, instructor (Admin only), attendance status, free-text search (student name, batch, instructor).	App
Student filter	Implemented as free-text search by student name, not a dedicated student picker. For an instructor, student-name search does not match (students are not visible to instructors outside a roster).	App, DB
Detail	Per-session roster from the frozen eligible list with each mark.	App
Edit	Edit → Review Changes → Confirm & Save.	App
Batch-scoped view	Batch Details → Attendance tab reuses this list for one batch.	App
Student-scoped view	Student Details → Recent Attendance.	App, Test

15.1 Historical integrity

History survives a student changing or leaving a batch, a membership expiring or being cancelled, a batch being deactivated, and a schedule changing, because both the session details and the eligible list are frozen when attendance is first saved. (*DB, Doc*)

16. Dashboard and Reports

16.1 Dashboard

Element	Admin	Instructor	Rule
Active Students	Yes	—	Count of students with stored status Active.
Active Batches	Yes	—	Count of batches with stored status Active.
Today's Classes	Yes	Yes	All occurrences today (stored + calculated); instructor sees own.
Attendance Marked	marked / total	"Completed" + "Remaining"	Counts sessions whose stored status is completed — not those merely displayed as Completed by the clock.
Today's class list	All	Own	Time, batch, instructor (Admin), counts, status, link to attendance.
Upcoming classes	Next 6 days	Next 6 days (own)	
Calendar	Yes	Yes (own)	Days with sessions; links to that day's sessions.
Recent Activity	Yes	—	Latest 4 of: attendance saved, student added, membership created — read from the records, no activity log.

No charts, trends or analytics are built. (*App*)

16.2 Reports (Admin only)

All three reports cover **completed sessions only**, use **inclusive** date ranges (both dates required), and read eligibility from the frozen lists. (*App, DB*)

Report	Filters	Shows
Student Attendance	Student, date range	Sessions the student was eligible for (or holds a mark for); Present, Absent, Unmarked; attendance % = present ÷ eligible sessions; per-session table.
Batch Attendance	Batch, date range	Sessions, eligible total, present, absent; Average Attendance = pooled present ÷ pooled eligible (not a mean of per-session %); per-session table.
Attendance Summary	Date or date range	Total sessions, present, absent, overall attendance % (pooled), per-session table across all batches.

Rule	Evidence
Unmarked is never counted as Absent.	App
When nothing is eligible, the percentage is blank / "—", not 0% (deliberately different from session screens).	App
Export: CSV and Excel (.xlsx) of the full filtered result, same figures as on screen.	App
Center profile details are not included in exports.	App
Google Sheets export / sync: not built .	App

17. Settings and Configuration

Tab	Content	Rules	Evidence
Center Profile	Name (required, ≤ 100), logo, address (≤ 300), phone (digits), email	Exactly one center profile can exist; it cannot be created or deleted from the app. The name and logo appear in the application shell and on receipts.	DB, App
Regional Settings (part of Center Profile)	Time zone (IANA, default Asia/Kolkata), currency (ISO 4217, default INR)	Invalid zones/codes rejected. Time zone governs every business date (Section 18). Changing currency never converts existing amounts; each membership keeps its own currency.	DB, App, Test
Instructors	See Section 7.		
Roles & Permissions	Read-only description of Admin and Instructor access; Change Password card for the signed-in Admin.	No role configuration.	App

Any signed-in user (including an Instructor) can read only the center name, logo, time zone and currency — not its address, phone or email. (DB)

18. Timezone and Business Date Rules

Rule	Evidence
The center's time zone (one setting, default Asia/Kolkata) defines "today" and "now" everywhere. It never depends on the user's browser or device.	DB, App, Test
Daylight-saving rules come from the IANA time zone database.	Test
Session, membership, enrollment, assignment and schedule dates are calendar dates (no time of day).	DB
Session Upcoming / Ongoing / Completed display compares the center's current date-time to the session's date + start/end time.	App, Test
Membership status uses the center's current date; both ends inclusive.	App, Test
Attendance eligibility compares dates to the session date , never today.	DB, Test
"Future session" for attendance means session date after the center's today.	DB
A membership cancellation timestamp is converted to a center calendar date to decide which sessions it affects.	DB
Schedule edit and schedule-change dates cannot be before the center's today.	App
"Unstarted" assignments (deletable) are those starting on or after the center's today.	DB
There is no per-user, per-batch or per-schedule time zone.	Doc

Minor exception: the "date of birth cannot be in the future" check uses the server's UTC date rather than the center date. (*App*)

19. Authorization and Security Rules

Rule	Evidence
Every page and every action checks the user's role on the server; hiding a menu item is not the security boundary.	App
Every data table is protected by row-level security: Admin rows for admins; instructor rows only for their own sessions, schedules, batches taught and attendance.	DB
Instructors have no direct write permission on any table. Their two writes (storing a session, saving attendance) go through database functions that check ownership.	DB
Signed-out visitors can read nothing.	DB
The application role can never be set by the user; new accounts are always Instructor.	DB
No record of a historical entity can be deleted by any role. The only deletes are: unused schedules (through a checked function) and unstarted schedule assignments.	DB
Sign-in errors and password-reset responses do not reveal whether an account exists.	App
Changing a password inside the app requires the current password. Minimum password length is 8.	App, Test
Photos and logos are publicly readable by URL but only admins can upload, replace or remove them.	DB

20. Core Business Workflows

20.1 Onboard a student (Admin)

Students → Add Student → Save
→ Add Membership (or Skip)
→ Add Batch Enrollment (batch + at least one schedule)
→ Student Details (completion)

20.2 Create a batch and its timetable (Admin)

Batches → Add Batch → Batch Details → Add Schedule
(days × time slots, instructor, effective from) → one schedule per combination

20.3 Take attendance (Instructor or Admin)

Dashboard / Attendance → Today's session → Take Attendance
→ mark Present / Absent (or Mark All Present) → Save
→ session Completed; eligible list frozen

20.4 Correct attendance

Attendance History → session → Edit → change marks → Review Changes → Confirm & Save

20.5 Change a recurring class (Admin)

Schedule → Edit
├ unused → corrected in place
└ used → choose effective date ($\geq$ today) → old version closed, new version from that date

20.6 Change one session (Admin)

Session Details → Edit This Session (Upcoming only) → instructor / time → Save
Session Details → Cancel / Holiday → optional note → Confirm (irreversible)

20.7 Change which classes a student attends (Admin)

Student Details → Edit Enrollment → change schedule selection
→ schedule change date ($\geq$ today) → removed: closed / withdrawn; added: start on that date

20.8 Renew / cancel a membership (Admin)

Membership Details → Renew → new membership (start = previous end + 1) → Save
Membership Details → Cancel → Review → Confirm (record kept, irreversible)

20.9 Give an instructor access (Admin)

Settings → Instructors → Add Instructor (with email, Active)
→ Provide Login Access → invitation email → instructor sets password → signs in
→ assign to schedules

21. Business Rules and Validations

Consolidated register. Section numbers point to the detail.

#	Rule	Where enforced	\$
R1	Only Admin and Instructor roles exist; new accounts are always Instructor.	DB	4
R2	Student ID <code>YC-000001</code> and Membership ID <code>MEM-000001</code> are generated, unique and immutable.	DB	6, 10
R3	Batch short code is required and unique (case-insensitive); stored upper-case.	DB, App	8
R4	Instructor email is unique when present; student phone/email are not unique.	DB	6, 7
R5	One active enrollment per student per batch.	DB	9
R6	An enrollment must have at least one schedule assignment, all from its own batch.	App	9
R7	Schedule-change dates and schedule edit dates cannot be before today.	App	9, 12
R8	No overlapping non-cancelled memberships per student.	DB	10
R9	Membership end $\geq$ start; amount > 0 ; payment defaults to Pending.	DB, App	10
R10	Membership status is derived; cancellation overrides; cancellation is one-way.	App, DB	10
R11	Payment status never affects eligibility.	App	10, 11
R12	Schedule end time $>$ start time; effective until $\geq$ effective from; every schedule has a batch and an instructor.	DB	12
R13	Overlapping schedules are allowed.	DB	12
R14	Used schedules are versioned or deactivated, never edited in place or deleted; unused schedules may be corrected or deleted.	DB, App	12
R15	At most one stored session per schedule per date; no ad-hoc sessions.	DB	13
R16	Only Upcoming sessions can be edited; only <code>scheduled</code> sessions can be cancelled / marked holiday; both irreversible for history.	App	13
R17	Stored status <code>completed</code> is set only by saving attendance.	DB	13, 14
R18	Eligibility = active student + active enrollment + schedule assignment + covering membership, on the session date.	DB	14
R19	No attendance for future, cancelled or holiday sessions.	DB	14
R20	Saving may leave students unmarked and still completes the session.	DB	14
R21	One attendance record per student per session; never deleted.	DB	14
R22	The eligible list is frozen on first save and never changes.	DB	14, 15
R23	Attendance % = Present $\div$ Eligible.	App, DB	14, 16
R24	One center profile; one time zone; one currency.	DB	17, 18
R25	Nothing historical is deleted.	DB	19

22. Data Relationships

Only relationships present in the V1 database are shown.

Sign-in account	-1:1-	Profile (role: admin instructor)
Sign-in account	-0..1:0..1-	Instructor
Student	-1:N-	Batch Enrollment
Batch Enrollment	-1:N-	Schedule Assignment
Schedule Assignment	-N:1-	Schedule Series
Schedule Series	-N:1-	Batch
Schedule Series	-1:N-	Schedule (version)
Schedule (version)	-N:1-	Instructor
Schedule (version)	-N:1-	Batch
Schedule (version)	-1:N-	Class Session (stored)
Class Session	-N:1-	Batch (snapshot)
Class Session	-N:1-	Instructor (snapshot)
Class Session	-1:N-	Attendance
Attendance	-N:1-	Student
Class Session	-1:N-	Frozen Eligible Student
Frozen Eligible Student	-N:1-	Student
Student	-1:N-	Membership (amount, currency, payment status)
Center Profile		(single record; not referenced by other records)

Relationship	Nature	Evidence
Membership ↔ Batch Enrollment	No stored link. Derived by date overlap for display.	DB
Payment	Not an entity. Attributes of Membership.	DB
Center ↔ anything	No link. Single implicit center.	DB
Deletion behaviour	Every link restricts deletion; nothing cascades.	DB

23. Edge Cases and Special Rules

23.1 Handled by V1

Case	V1 behaviour	Evidence
Student moves to another batch on the same day as a recorded session	The recorded session keeps its frozen roster and marks.	DB
Membership cancelled after a session happened	Session stays eligible; only sessions on/after the cancellation day are affected.	DB, Test
Membership ends today vs. tomorrow across midnight UTC	Status flips at the center's midnight, not UTC midnight.	Test
Two people save attendance for the same session at once	Serialized by a lock; no duplicate records.	DB
Two people store the same session at once	Second request receives the first one's session.	DB
Schedule deleted while someone records a session	The delete re-checks and refuses.	DB
Session reassigned to another instructor	Access moves to the new instructor.	DB
Zero eligible students	Save allowed; session completes; 0% on session screens, "—" in reports.	DB, App
Mark for a student no longer in the eligible list	Kept as evidence; unmarked count floors at zero.	DB
Removing a schedule assignment that hasn't started	Withdrawn, not left as a one-day assignment.	DB, App
Centre currency changed	Old memberships keep their currency; renewal leaves amount blank if currencies differ.	App
Legacy phone numbers without a country code	Kept as stored; only unambiguous Indian mobile numbers were given +91 .	DB
Multi-step saves that partly fail (enrollment + assignments, schedule close + new version)	Schedule versioning is rolled back on failure; an enrollment left with no assignment is flagged on screen.	App

23.2 Sessions completed before the frozen-list feature

Sessions completed before migration 0015 were back-filled where evidence existed. Any session with no frozen list still resolves eligibility live and is labelled as such internally. Eligible-but-unmarked students lost from such sessions are accepted as unrecoverable. (DB, Doc)

23.3 Current-status effects on unrecorded dates

Student status and enrollment status are **not date-ranged**. Deactivating a student or an enrollment therefore removes them from the eligible list of **every session not yet recorded — including past dates nobody recorded**. Recorded sessions are protected by their frozen list. (DB)

23.4 Deactivations do not cascade

Deactivated	Not changed automatically
Student	Enrollments, assignments, memberships
Batch	Its schedules (sessions keep being calculated), enrollments
Instructor	Their schedules (sessions keep being calculated); only an Admin can then record those sessions

Deactivated	Not changed automatically
Enrollment	Its schedule assignments (but eligibility stops because status is checked)

(App, DB)

23.5 Other noteworthy behaviour

- Enrollment start/end dates can be edited freely, including into the past. *(App)*
 - Schedule deactivation's effective-until date may be set in the past (only bounded by effective from), whereas schedule edits cannot be back-dated. *(App)*
 - Cancelled memberships can still be edited and renewed. *(App)*
 - An instructor can be invited only once; an existing account with the same email blocks the invitation. *(App)*
-

24. V1 Limitations / Out of Scope

Not supported in V1	Evidence
Multiple centers / branches; any center-level data separation	DB, Doc
Student login or self-service	Doc, DB
Roles other than Admin / Instructor; custom roles; role configuration	DB, Doc
Admin creation from the UI	DB
Revoking an instructor's login (other than deactivation)	App
Assigned Classes screen	App
Payment records, online payments, automatic billing, reminders, invoices, tax	DB, Doc
Revenue, retention or advanced analytics; charts	Doc, App
Batch capacity	Doc
Online / offline attendance mode; statuses other than Present / Absent	Doc, DB
QR, face recognition, mobile app, AI features	Doc
WhatsApp / SMS / email integrations (only a click-to-chat link on receipts)	Doc, App
Google Sheets export or sync	App
Center details in exported reports	App
Ad-hoc (non-scheduled) sessions	Doc, DB
Un-cancelling a session or a membership; reactivating a deactivated schedule	App
Adding a student to a completed session's frozen roster	DB, Doc
Audit log of who changed what	DB
Per-user / per-schedule time zone	Doc

25. V1 Business Architecture Summary

Layer	V1 position
Tenancy	One implicit center; single settings record.
Actors	Admin (everything), Instructor (own sessions' attendance).
Master data	Students, Instructors, Batches — deactivated, never deleted.
Commercial	Membership periods with amount, currency, paid/pending; no payment entity.
Membership in classes	Batch enrollment + schedule assignment, date-ranged.
Timetable	Versioned recurring schedules grouped into stable series.
Operations	Sessions calculated from the timetable, stored on first touch with a snapshot.
Attendance	Present/Absent against a computed, then frozen, eligible list; atomic save.
Insight	Dashboard counts, attendance history, three reports with CSV/Excel.
Time	One center time zone defines every business date.
Integrity	Database-enforced: no deletion of history, frozen rosters, snapshots, versioning.

Audit and history behaviour (summary)

V1 preserves **state history**, not an **action log**:

Preserved	How
Past enrollments and assignments	Closed with end dates, never overwritten.
Past schedule versions	Closed and superseded, never deleted.
Past memberships	Renewal creates a new record; cancellation is a timestamp.
Session details	Snapshot at first touch.
Session roster	Frozen eligible list at first save, with provenance (save vs. back-fill).
Attendance	Updated in place on correction — the previous value is not kept .
Who made a change	Not recorded (only created/updated timestamps).

V1 → V2 Transition Notes

This section does **not** design V2. It only identifies how V1 concepts relate to a future multi-center product.

A. Single-center assumptions in V1

Assumption	Where it lives
Exactly one center record can exist.	Center profile (singleton)
No record carries a center reference; "belongs to the center" is implicit.	Every table
One time zone for all business dates.	Center settings, eligibility, session status, membership status
One currency for new memberships.	Center settings
Student ID and Membership ID sequences are global.	ID generation
Batch short codes are unique across the whole system.	Batch rule R3
Instructor emails are unique across the whole system.	Instructor rule R4
"Admin" means admin of everything; there is no scoped admin.	Roles and data permissions
An instructor account maps to exactly one instructor record.	Instructor identity
One active enrollment per student per batch, and no overlapping memberships per student, are evaluated system-wide.	R5, R8
Dashboard counts, reports and history cover all data.	Dashboard, Reports, History
Logo, photos and batch images share one storage area.	Storage

B. V1 concepts likely to be affected

Concept	Why it is likely affected
Center Profile / Regional Settings	Currently a single record holding the only time zone and currency.
Roles and permissions	Two global roles; data permissions are "admin sees all".
Student identity and Student ID	Students and their IDs are global to the one center.
Membership	Overlap rule, currency and ID sequence are system-wide.
Batch	Short-code uniqueness is system-wide.
Instructor	Email uniqueness and one-account-one-instructor link are system-wide.
Time-zone-dependent rules	"Today", session status, eligibility cut-offs and schedule date checks all read the one center time zone.
Dashboard, Reports, Attendance History	Aggregate across all data with no center dimension.
Settings → Roles & Permissions	Describes the fixed two-role model.

C. V1 concepts that appear reusable without change

These are defined without reference to a center and behave the same regardless of how many centers exist (subject to V2 decisions):

Concept
Separation of Student, Membership, Batch, Enrollment, Schedule Assignment, Schedule, Session, Attendance

Concept
Four-condition eligibility evaluated on the session date
Schedule series + versioning; unused-schedule correction/deletion rule
Projection and materialize-on-first-touch of sessions, with snapshot
Stored vs. displayed session status
Atomic attendance save; Present/Absent; unmarked not stored; completion on save
Frozen eligible list per session
Derived membership status; renewal as a new record; one-way cancellation
Date-ranged enrollments and schedule assignments; close-never-rewrite; withdraw-unstarted
"Deactivate, never delete" for historical entities
Report counting rules (pooled ratio, unmarked ≠ absent, completed sessions only)
Per-membership currency stamping

Appendix A — Documentation vs Implementation Discrepancies

Resolved in favour of implemented behaviour, as instructed.

#	Topic	Documentation says	V1 implementation	Resolution in this document
A1	Phase status	04-development-plan.md lists Phases 16, 16A, 17, 18, 19 as "In progress / not committed".	Reports, Dashboard, Settings, regional settings are committed and in use.	Treated as implemented; plan table is stale.
A2	Enrollment date protection	Plan Phase 16A step 6: enrollment / schedule-change dates should not conflict with existing attendance — "not yet built".	Not built. Enrollment dates are freely editable; schedule change only blocks dates before today. Recorded sessions are protected by frozen rosters instead.	Documented as current behaviour (§9, §23.5).
A3	Assigned Classes	UX IA lists it for Instructors.	No screen; navigation entry hidden from everyone.	Listed as not built.
A4	Center info in exports	Product §11: "can also be used in exported reports".	Not used in exports (used in app shell and receipts).	Listed as not supported.
A5	Inactive student "from that point on"	Product §8: excluded from new eligibility from that point on.	Status is not date-ranged; unrecorded past sessions also lose the student.	Documented (§23.3). Not confirmed as intended.
A6	"Active Batches"	Dashboard and Batches list both show "Active Batches".	Dashboard counts stored status; Batches list counts derived status (requires a current schedule). The numbers can differ.	Documented both (§8, §16.1).
A7	"Active Students" caption	Dashboard tile caption: "Currently enrolled".	Counts students with Active status regardless of enrollment.	Documented actual rule.
A8	Cancelled membership actions	Product: cancellation is final.	Edit and Renew remain available on cancelled memberships.	Documented; intent Not confirmed .
A9	Schedule deactivation date	Product: changes apply to future sessions only.	Deactivation's effective-until can be a past date (edits cannot).	Documented; intent Not confirmed .
A10	Batch / instructor deactivation	Product does not state effect on schedules.	Schedules continue; sessions keep appearing.	Documented; intent Not confirmed .
A11	Inactive instructor experience	Code comments anticipate an "access disabled" screen.	Not built; instructor sees empty screens.	Documented (§5.3).
A12	"At least one schedule assignment"	Product rule; plan asks whether app-level enforcement is acceptable (open item).	App-level only; partial failures flagged on screen.	Documented (§9).
A13	Attendance History "Student" filter	Product §9 lists a Student filter.	Implemented as free-text search; not effective for instructors.	Documented (§15).
A14	Google Sheets	Product §10: "may be supported".	Not built.	Listed as not supported.

#	Topic	Documentation says	V1 implementation	Resolution in this document
A15	Zero-eligible percentage	Product §8: 0%.	0% on session screens; blank/"—" in reports (deliberate).	Documented both (§14.5, §16.2).

Appendix B — Evidence Sources

Source	Used for
supabase/migrations/0001 – 0024	Entities, constraints, row-level security, eligibility, attendance save, snapshot, schedule deletion, regional settings
lib/*/actions.js , lib/*/validation.js , lib/*/data.js	Server-side rules, validations, derived statuses, reports
app/** route guards and pages	Role access, screen-level behaviour
lib/**/* .test.js (370 tests, all passing)	Time zone, membership validity, schedule usage, batch summary, session editability, settings validation
supabase/verification/*	Manual database verification scripts (not run as part of this audit)
docs/01-product.md , docs/02-ux.md , docs/04-development-plan.md	Intended rules, flows, open items